



Rapid Neutralization of chemical warfare agents

10-100x FASTER DETOX OF NERVE AGENTS WITH REUSABLE CATALYTIC MATERIALS

Current systems either trap toxins or degrade them too slowly-leaving users exposed

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1 THE PROBLEM

Nerve agents (GB, VX) are among the most dangerous chemical threats.

- Fast-acting & lethal**
Effective at ppm levels.
- Current systems fall short**
Either capture only (saturation risk) or degrade too slowly.
- Operational gap**
Need rapid, continuous, field-deployable neutralization.



2 VALUE PROPOSITION

SPEED

10–100× faster hydrolysis vs. leading MOFs

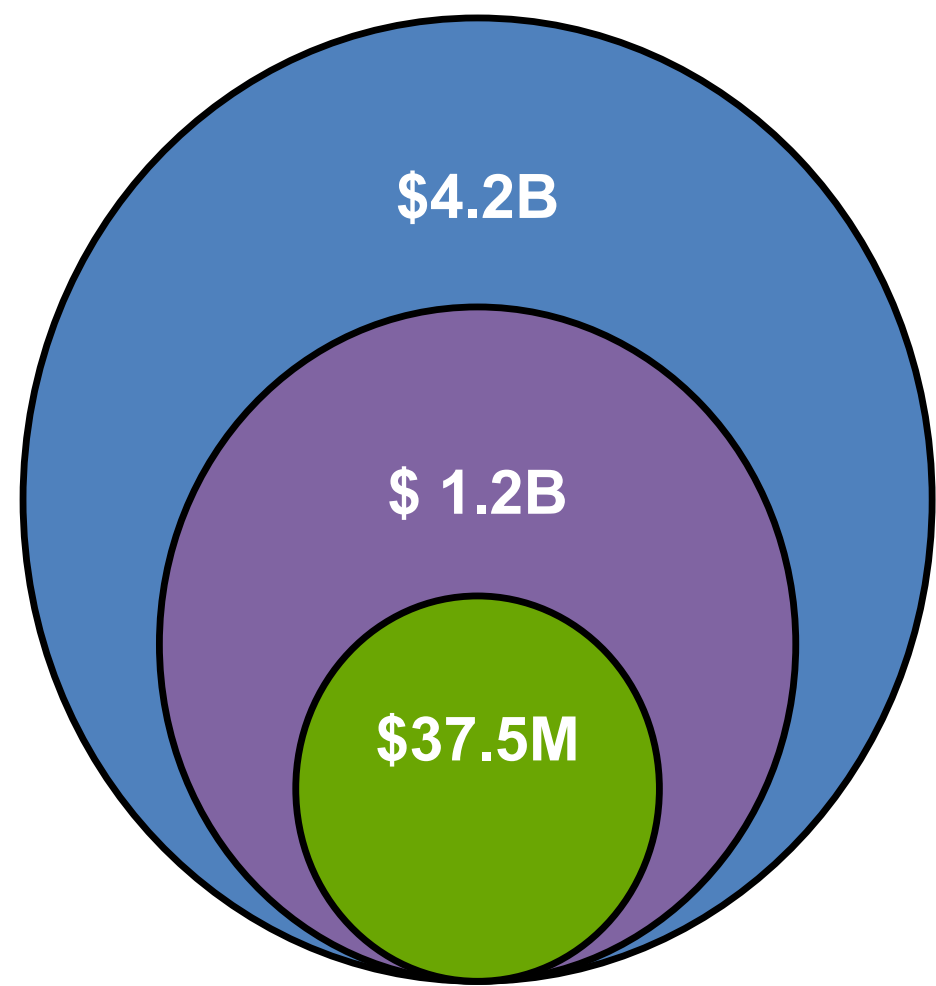
COMPLETENESS

Near complete degradation at ~10 ppm concentration

REUSABILITY

Sustained catalytic activity over 8+ hours of continuous-flow operation

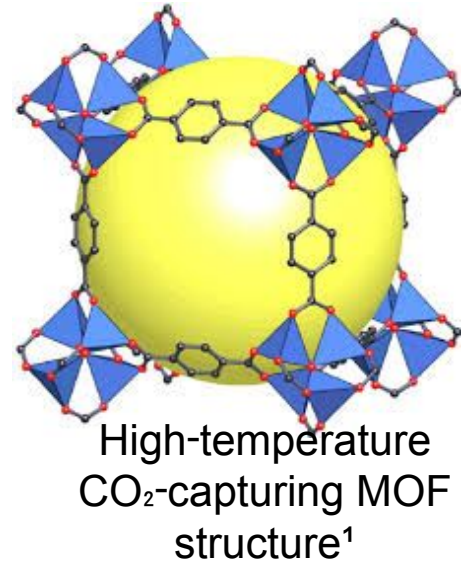
3 MARKET POTENTIAL



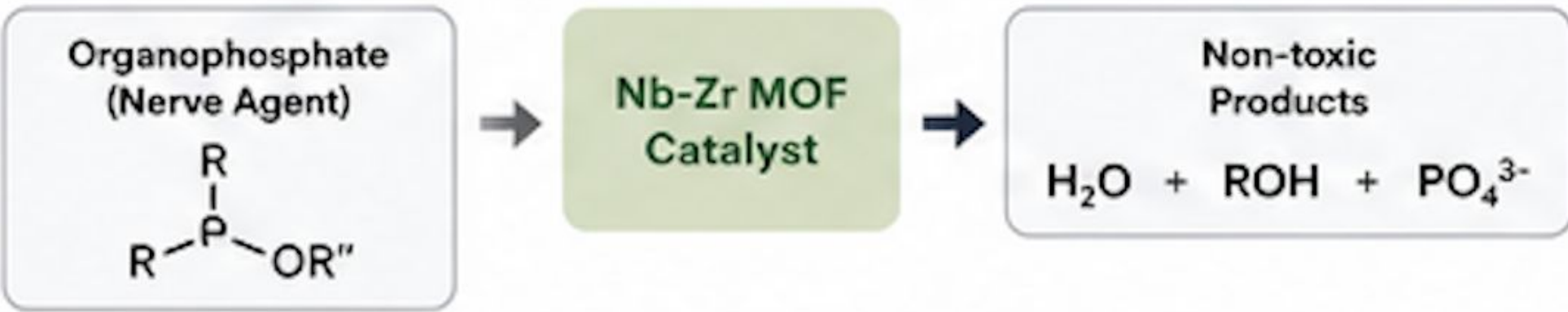
- Total Addressable Market (TAM)**
3.2-4.2 Billion USD, 6.4% CAGR through 2033
- Serviceable Addressable Market (SAM)**
Total deployable CBRN filtration + protective materials market (~\$6B) × MOF adoption rate (~20%) = \$1.2 Billion SAM
- Share of Market (SOM)**
50 organizations × 500 units × \$1,500 per unit = \$37.5 Million

4 TECHNOLOGY OVERVIEW

A novel Niobium-substituted Zirconium Metal-Organic Framework (Nb-Zr MOF) that catalytically breaks down organophosphates into non-toxic products.



- High activity at low concentrations (ppm)
- Reusable solid catalyst (not consumed)
- Compatible with filters and flow systems



5 COMPETITIVE LANDSCAPE

Category	Key Players / Groups	Core MOF Systems	Application Focus	Relevance to Nb-Zr MOF
Academic leaders	Northwestern, University of Oslo	UiO-66, NU-1000, MOF-808	CWAs, gas storage, catalysis	Direct competition
Defense research	DARPA, US Army, DTRA	NU-1000, UiO	Nerve agents, filtration	Close use case
Water MOF groups	China, EU, US	UiO-66-NH ₂	PFAS, heavy metals, uranium capture	Adjacent
Catalysis MOF groups	EU, US, China	MOF-808, UiO	Biomass conversion, acid catalysis	Reaction engineering
Commercial MOF companies	NuMat, Svante	Custom MOFs	Gas storage, semiconductors, carbon capture	Platform competitors
Filtration / PPE	contractors, startups	Zr-MOF coatings, fabrics	Protective gear, filters	downstream competition

6 BUSINESS MODEL & GO-TO-MARKET

BUSINESS MODEL

- License our MOF technology to filter & PPE manufacturers
- Revenue from material supply + licensing/IP
- High-value applications in defense, industrial safety, and water treatment

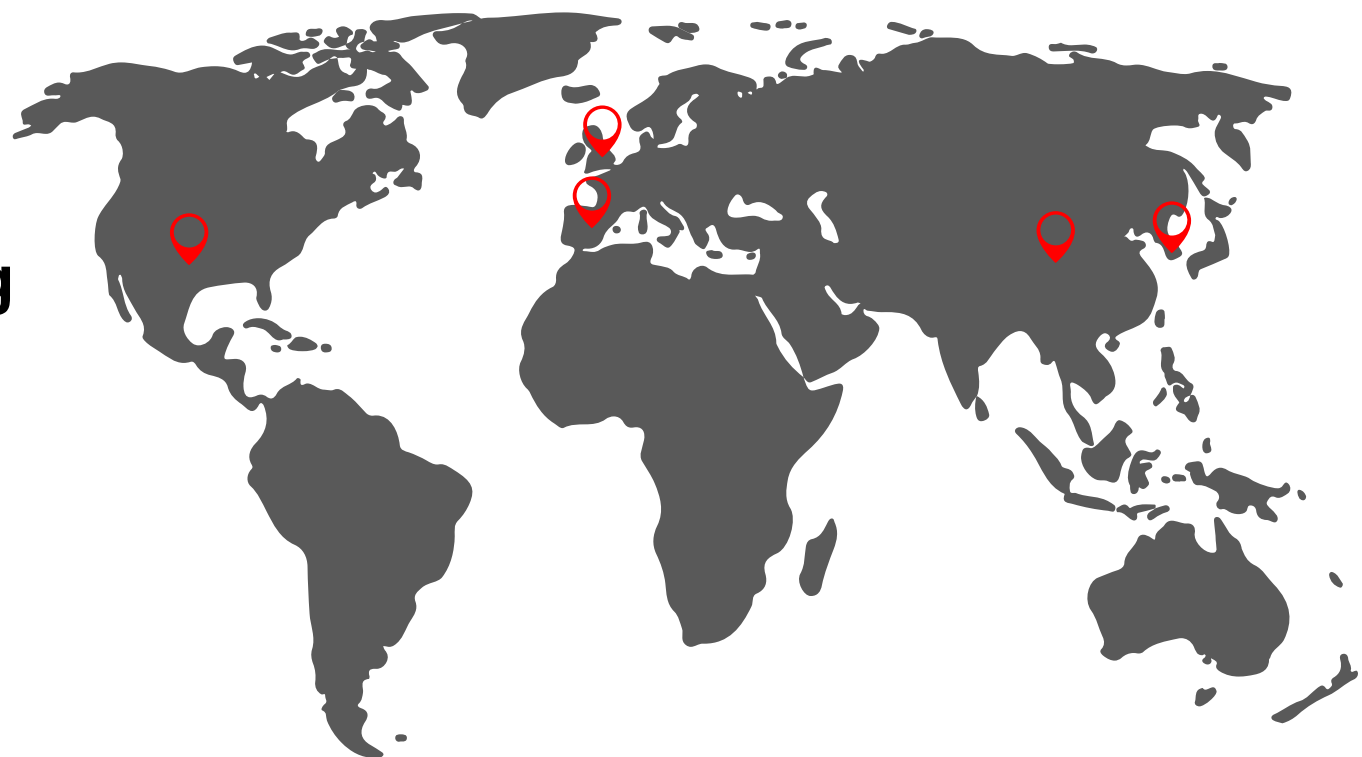
GO-TO-MARKET STRATEGY

- Begin by licensing technology to Numat to develop PPE technologies
- Licensing Agreements between LBNL and companies who desire to use the specific Nb MOF technology
- Filing for patents in other countries with extensive MOF research allows the technology to stay in the United States

7 IP ANALYSIS

Based on market research in MOF production

- Patent pending in the United States and looking to patent globally
 - China (CNIPA)
 - Korea (KIPO)
 - Spain (EPO)
 - UK (IPO)



8 SCALING

PRODUCTION STAGE

1.		Lab/Current (retail price)	~\$150/g = \$150,000/kg
2.		Pilot scale (~1 ton/year)	~\$500-2000/kg
3.		Industrial scale (~100 ton/year)	~\$40-70/kg (\$5-10 more than Zr-Mofs)
4.		Optimistic large scale (>1kton/year)	~\$15-20/kg (competes with alternatives)

8 ECONOMICS

CANISTER FILTER (Example)

MOF Amount	10-20 g
MOF Cost per Unit (At Scale)	~\$1
Cost Increase vs. Zr-MOF	~\$0.20

AVON GPCF50 (NATO CBRN filter 40 mm)

Cost of Goods Sold	~\$69
MOF % of Cost	~1.5%
Switching Cost	Low